Module 3 Homework 4 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:15

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

Suppose we have these four equations:

$$y = 3x + 2$$
$$y = 3x + 5$$
$$y = -x + 10$$
$$y = -x + 15$$

Consider the points that we get from these lines crossing each other. If you connect all of these points with straight lines, what kind of shape do you get?

- parallelogram
- triangle
- circle
- square
- line

2

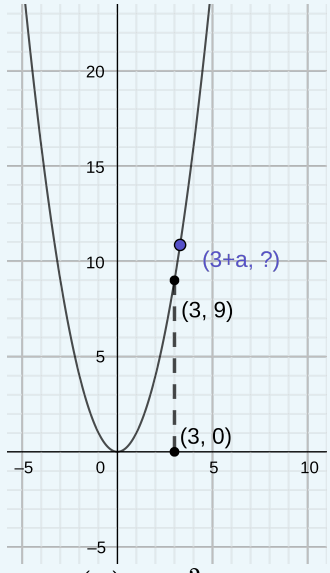
Review Later

In order to find the point E that is equidistant (equal distance) from three points, A, B, and C, you would do which of the following steps?

- Connect the midpoints of $\overline{AB}$ and $\overline{BC}$, and E is the midpoint of this line.
- Find the midpoint of the line connecting A and C. This is E.
- Draw an arc with B as the center, passing through A and C, and point E is at the middle of this arc.
- Draw B as the equidistant point.
- Construct a line perpendicular to $\overline{AB}$ passing through its center, and construct a line perpendicular to $\overline{BC}$ passing through its center. E is where these two lines cross.

3

Review Later

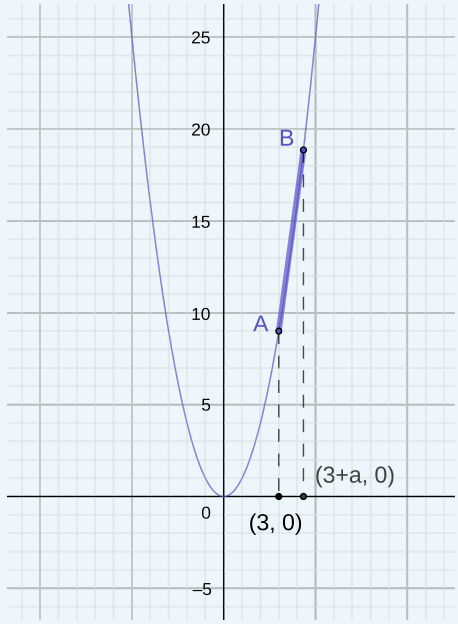


An ant starts walking along a parabola, $y(x) = x^2$, starting from the point where $x = 3$. He walks until a point where $x = 3 + a$, where a is a positive number. What is the y-coordinate of the point that he is at now?

- $6 + 2a$
- $9 + 6a + a^2$
- $9 + a^2$
- $9a$
- $9 + a$

4

Review Later



An ant walks along a parabola, $y = x^2$, from point A, on the curve where $x = 3$, to point B, on the curve where $x = 3 + a$, and a is positive. Find the slope of the straight line $\overline{AB}$.

- 6
- 3
- 2
- $6 + a$
- $3 + a$

5

Review Later

If the distinct lines $y = ax + b$ and $y = -7x + 49$ never intersect, what is the value of a?

6

Review Later

Which of the following choices corresponds to equation(s) of a straight line?

- $y = 1$

- $y = x + 1$
- $y = x^2 + 2x + 1$
- $y = x^3 + 3x^2 + 3x + 1$
- The first and second choices

7

Review Later

Find the perimeter of a triangle bounded by a straight line $5x + 12y = -60$ and coordinate axes.

8

Review Later

You are looking for a new phone plan and found two options: Pi-Phone and Euler-Mobile. Pi-Phone costs $\$30$ a month plus 5 cents for every call you make. Euler-Mobile, on the other hand, only costs $\$3$ a month, but calls cost 16 cents. How many calls do you need to make for Pi-Phone to be a better deal?

9

Review Later

There are six different parallel lines of slope $\frac{1}{3}$ that are equal distance apart. If the distance between the y-intercepts of two adjacent lines is 72, find the distance between the x-intercepts of those two adjacent lines.

10

Review Later

What is the slope of the line that touches the graph of the curve $y = x^2 + x$ at only the point (1, 2)?

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

Points $A(1,1)$, $B(4,2)$ and $C(0,7)$ are the vertices of the triangle ABC. What is the slope of the line that contains the height of the triangle drawn to AB?

12

Review Later

Find the slope of the line containing the midpoint of $(0,3)$ and $(7,8)$ and the midpoint of $(-3,-19)$ and $(4,10)$. Your answer should be a fraction in simplest terms, with no spaces, and not a mixed number, e.g. 3/2

13

Review Later

The sum of the x-coordinates of a triangle's vertices equals $\sqrt{71}$. What is the square of the sum of the x-coordinates of the midpoints of the sides of this triangle?

14

Review Later

A square has opposite corners at $(11,2)$ and $(15,10)$. Find the sum of the x-coordinates of its other two corners.

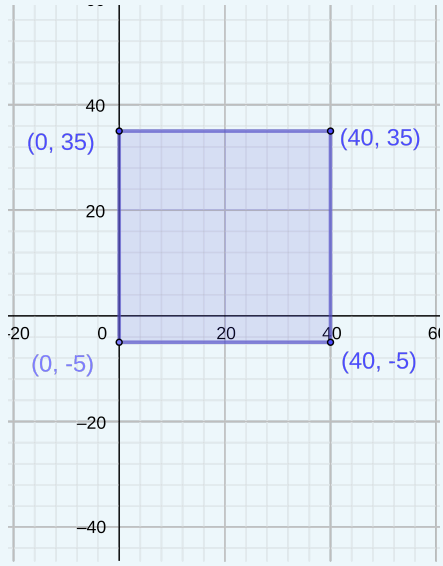
15

Review Later

A banana is at $(0,0)$. A monkey is running along $x+2y=20$ towards the banana. At some point (a,b) the distance from the monkey to the banana stops decreasing and starts to increase. What is $a+b$?

16

Review Later



A lattice point is a coordinate (x, y) where both values of x and y are integers. The graph of $3y = 2x + 15$ crosses how many lattice points inside or on the border of the shaded square region above?

17

Review Later

Ten different lines pass through the point $(3, 4)$, and they each have different slopes which form an arithmetic progression, starting from a slope of $-1, -2, -3, -4, -5, -6, -7, -8$ and -9 to a slope of -10. What is the sum of the y-intercepts of these ten lines?

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

The length of the straight line between $(3, 7)$ and $(5x - 6, 6x - 5)$ is $\sqrt{193x}$. What is the sum of the possible values of x?

19

Review Later

Squares A and B both have dimensions 2019×2019. Square A has both its dimensions enlarged by integer x, while Square B has both its dimensions decreased by this same quantity x. What is the least x such that the resulting difference between the area of Square A and the area of Square B is at least 2019×2019?

20

Review Later

A helicopter flies to a point 2 km north and 1 km west of its starting point. Then it turns to the right at a 90° angle and flies a distance equal to its previous distance plus $\frac{1}{2}$ of **its very first distance.** (Note: Not half of its previous distance, but half of its first leg.) How many kilometers away will the helicopter be from its original position after 96 such turns?

Your answer should be in the form $a\sqrt{b}$, where b does not contain any square factors.

$\sqrt{}$

View Marked Questions

Finished

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